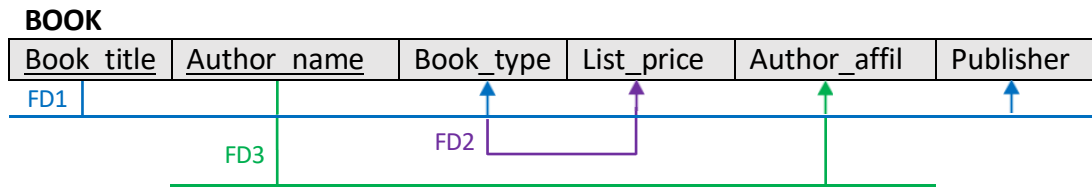


CSC430/530 – Database Management Systems

Assignment #4 – Functional Dependencies & Normalization

GAVIN DOMINIQUE

1. Consider following relation for published books:



- *Author_affil* refers to the affiliation of the author.
- Primary key is {*Book_title*, *Author_name*}.
- Functional dependencies are:
 - FD1: *Book_title* → *Publisher*, *Book_type*
 - FD2: *Book_type* → *List_price*
 - FD3: *Author_name* → *Author_affil*

a. What normal form this relation in (1NF, 2NF, 3NF)? Justify your answer by describing violations of normal forms (if any).

It is part of 1NF since it violates 2NF and 3NF by having partial and transitive dependencies.

b. Describe steps to normalize this relation up to 3NF. For full points, show all decomposed relations.

The first step is to convert to 2NF, this is done by creating multiple tables to get rid of partial dependencies:

BOOK

<u>Book_title</u>	Book_type	Publisher
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Author

<u>Author_name</u>	Author_affil
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Written_By

<u>Book_title</u>	List_price	<u>Author_name</u>
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The next step is to convert from 2NF to 3NF, this is done by getting rid of transitive dependencies:

BOOK

<u>Book_title</u>	Book_type	Publisher
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Author

<u>Author_name</u>	Author_affil
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Written_By

<u>Book_title</u>	<u>Author_name</u>
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Book_Info

<u>Book_type</u>	List_price
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2. Define which of the provided functional dependencies may hold for the given relation. If the dependency does not hold, explain why by specifying tuples that cause the violation.

	Instructor	Course	Text	Quarter	
1	Smith	Data Structures	Bartam	Spring	A. Text -> Course
2	Hall	Systems Programming	White	Winter	B. Text -> Instructor
3	Brown	Programming Languages	Williams	Summer	C. Instructor -> Course
4	Smith	Data Structures	Bartam	Winter	D. Course -> Text
5	Ross	Data Mining	Williams	Summer	E. Course -> Quarter
6	Hall	Systems Programming	White	Spring	
7	Johnson	Databases	Elmasri	Fall	

- A. Does not hold because the text Williams is connected to two different courses.**
- B. Does not hold because the text Williams is connected to two different instructors.**
- C. Holds**
- D. Holds**
- E. Does not hold because the courses Data Structures and Systems Programming are offered in Spring and Winter and Winter and Spring respectively.**